B. TECH-ME-III-I Sem

23A03503	METROLOGY MEASUREMENTS	L	T	P	C
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	Course objectives: The objectives of the course are to
1	Explain the system of limits, fits &tolerances and design of gauges.
2	Identify the use of flatness and surface gauges
3	Know the measurement of screw thread, Gear profiles.
4	Describe the Measurement of Displacement and Strain.
5	Illustrate the measuring process of Pressure, Force and Torque.

Course Outcomes: On successful completion of the course, the student will be able to,		
CO1	Demonstrate the concept of different types of dimensional tolerances and chose the desire limits and fit component to solve the required fit.	L2, L3,L6
CO2	Explain the basic standards of measurements and also apply the desired flatness and surface gauges to analyze the dimensions.	L2, L3,L4
CO3	Evaluate engineering parts with various precision instruments and choose the required surface roughness instrument to compare the parts.	L4,L5,L6
CO4	List out various measuring techniques for Displacement and Strain. Define the various instruments for measuring the displacement and calibrate the strain.	L2, L1,L5
CO5	Estimate the Instruments accuracy and Perform calibration of Force, Torque and pressure measuring instruments	L6, L5,L1

UNIT I

Concept of measurement

Concept of Measurement: Concept of feedback Control systems -generalized measurement system, units and standards, measuring instruments, sensitivity, readability, range of accuracy, precision, static and dynamic response, repeatability, systematic and random errors, correction, calibration, terminology and limits fits and tolerances, hole basis and shaft basis system, interchangeability.

Limit Gauges And Gauge Design: Plug, Ring, Snap, Gap, Taper gauges. Taylor's principle. Design of Go and No Go gauges.

Linear and Angular Measurement: Linear measuring instruments: Vernier instruments, micrometers, slip gauges, tool makers microscope. Comparators: Mechanical, pneumatic and electrical. **Angular measurements:** Sine bar, bevel protractor and angle dekkor, rollers and spheres used to determine the tapers.

UNIT – II

Flatness and Surface Roughness measurement

Flatness Measurement: Measurement of flatness – straight edges – surface plates, optical flat and autocollimators, interferometers and their applications.

Surface Roughness Measurement: Terminology systems, differences between surface roughness and surface waviness- Numerical assessment of surface finish - CLA, R.M.S Value-Ra , Rz values, Methods of measurement of surface finish-profilograph, talysurf, BIS symbols for indication of surface roughness.

UNIT – III

Screw Thread and Gear Measurement

Screw thread measurements: Elements of threads, errors in screw threads, various methods for measuring external and internal screw threads, screw thread gauges.

Gear Measurement: Gear tooth terminology, measurement of gear elements-run out, lead, pitch backlash, profile, pressure angle, tooth thickness, diameter of gear, constant chord and base tangent method.

Coordinate Measuring Machine (CMM)- Construction and features.

UNIT – IV

Measurement of Displacement and Strain

Measurement of Displacement: Theory and construction of various transducers to measure displacement – Piezo-electric, inductive, capacitance, resistance, ionization and photoelectric transducers, calibration procedures.

Measurements of Strain: Various types of electrical strain gauges, gauge factor, method of usage of resistance strain gauge for bending, compressive and tensile strains, usage for measuring torque, strain gauge rosettes.

UNIT – V

Measurement of Force, Torque and Pressure

Measurement of Force: Direct method - analytical balance, platform balance; elastic members – load cells, cantilever beams and proving rings.

Measurement of Torque: Torsion bar dynamometer, servo controlled dynamometer and absorption dynamometer.

Measurement of Pressure: Standards and calibration, basic methods of pressure measurement, dead weight gauges and manometers, High and low pressure measurement, Elastic transducers.

Textbooks:

1. Beckwith, Marangoni, Linehard, Mechanical Measurements, 6/e, PHI, 2013.
2. R.K. Jain, Engineering Metrology, 20/e, Khanna Publishers, 2013.

Reference Books:

1. Mahajan, Engineering Metrology, 2/e, Dhanpat Rai, 2013.
2. S.Bhaskar, Basic Principles - Measurments and Control Systems, Anuradha Publications, 2014.
3. Anand K Bewoor& Vinay A Kulkarni, Metrology & Measurement, 15/e, McGrawHill, 2015.
4. D.S. Kumar, Mechanical Measurements & Control, Metropolitan Publishers, 5/e, 2015.

Online Learning Resources:

- https://nitsri.ac.in/Department/Mechanical%20Engineering/MEC_405_Book_2,_for_Unit_2B.pdf
- https://www.digimat.in/nptel/courses/video/112104250/L47.html
- https:/
- www.digimat.in/nptel/courses/video/112106138/L01.html

https://www.digimat.in/nptel/courses/video/112106179/L01

- html https://www.youtube.com/watch?v=tczyyM4Dykc
- https://www.youtube.com/watch?v=_UsAiZmRC1M
- https://www.youtube.com/watch?v=oCkaxI19X8